

ABDELRAHMAN SAEED

MACHINE LEARNING ENGINEER

CONTACT

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PROFESSIONAL SUMMARY

Machine Learning Engineer with a strong software development background in Python, C#, and C++. Experienced in building end-to-end machine learning pipelines including data cleaning, preprocessing, feature engineering, model training, evaluation, optimization, and deployment. Improved decision accuracy by 30% and increased model performance by 25–40% across classification and regression tasks. Integrated machine learning solutions with scalable backend systems, improving production readiness by 35% and enabling reliable, maintainable deployment in real-world applications.



TECHNICAL PROJECTS

Deep Learning

Personal Project – 2025

Brain Tumor Classification

- data exploration and visualization of MRI images to identify visual differences between brain gliomas and meningiomas, as well as to analyze image size and pixel density.
- applying intensive data amplification techniques to improve model robustness and generalizability.
- applied a custom CNN, ResNet50, and VGG16 using transfer learning comparing their performance through accuracy, recall, F1 score, confusion matrices, fine-tuning analysis, and training curves to identify the most effective architecture.

Machine Learning

Personal Project – 2025

E-Commerce Sales Prediction

- Performed in-depth exploratory data analysis and visualization on a 500,000-transaction UK e-commerce dataset , uncovered seasonal sales trends like top-performing products.
- Performed in-depth exploratory data analysis and feature engineering, including cleaning (missing values, outliers, duplicates), temporal feature extraction, target variable creation, and feature scaling to prepare high-quality data for modeling.
- Built, trained, and compared six regression models to predict total sales; evaluated performance using MSE, MAE, and R² score, conducted hyperparameter tuning, and identified the best-performing model for accurate e-commerce sales forecasting.

TECHNICAL SKILLS

Programming Languages:

- C++, Python, C#, SQL
- Excel , Dart

Frameworks & Tools:

- Sklearn , Plotly , NumPy
- Pandas , TensorFlow
- Matplotlib , Keras , Streamlit
- Git , GitHub

Databases:

- SQLite , MySQL , PostgreSQL
- Microsoft SQL Server

Architecture:

- RESTful API Design
- CRUD
- REST inference endpoints

LANGUAGES

- Arabic – Native
- English – Intermediate (B1)

CERTIFICATIONS & COURSES

Artificial Intelligence

ALX Academy

2024

Machine Learning

Mindset Training

2025

Deep Learning

Mindset Training

2025

AI & Machine Learning Foundations

Sprints

2025

HCIA-Big Data

October 6 University

2025

SOFT SKILLS

Agile

Scrum

Strong problem-solving

effective communication

deep critical thinking

exceptional patience in research and information gathering

EDUCATION

Bachelor of Computer Science

Al-Ahram Canadian University

GPA(3.66) 2022 - 2026 (Expected)

Machine Learning

Personal Project - 2025

Cardiovascular Disease Classification

- Built and deployed an end-to-end machine learning pipeline on 70,000+ medical records, performing data cleaning, enforcing medical constraints, and engineering clinically meaningful features (BMI, pulse pressure) to improve data reliability and interpretability.
- Trained, tuned, and compared multiple classification models (Logistic Regression, KNN, SVM, Random Forest, Gradient Boosting, XGBoost) using cross-validation and GridSearchCV, identifying XGBoost as the best-performing model based on ROC-AUC, precision, recall, and F1-score.
- Saved the final optimized model using the pickle library and deployed it as an interactive Streamlit web application, enabling real-time cardiovascular disease prediction.

WORK EXPERIENCE

Web Developer / Bot Developer

NOV 11, 2024 - MAR 22, 2025

WhatsApp Customer Service Bot

- Developed a WhatsApp bot integrated with Meta's Business API to streamline customer service interactions
- Built a dynamic admin panel enabling non-technical users to manage categories, issues, and solution links without modifying the codebase
- Designed and implemented RESTful APIs using Flask
- Engineered the backend with SQLite for lightweight and efficient data management
- Developed a responsive front end using HTML, CSS, and JavaScript to enhance usability for admin users
- Deployed the complete system on Railway cloud hosting, ensuring reliable availability, environment configuration, and real-world usage.

Software Developer - Desktop Application

July 12, 2024 - July 30, 2024

Store Management System

- Designed and developed a desktop application for comprehensive store management, including invoice tracking, inventory input/output, and salary calculations
- Integrated accounting features to ensure full operational and financial control within a single platform
- Utilized .NET Framework for robust desktop application development
- Managed database operations using SQL Express and LINQ for efficient data handling and queries

Software Developer - Desktop Application

Sep 29, 2023 - Oct 15, 2023

Employee Management System

- Developed an HR system in Excel using VBA to manage employee data and roles efficiently
- Enabled role-based access control for secure data editing
- Delivered an easy-to-use solution that automated HR functions and reduced errors